

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES

SATURDAY, 18 MAY 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) questions in Section B, marks will only be awarded for the FIRST questions in that section in the order the candidate submits the answers.

No calculator is allowed for this paper.

Answer questions only in the answer booklet provided.

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**SECTION A**

- Q1. (a) The JavaFX and Event Handling questions.
- (i) Describe the lifecycle of a JavaFX application. (6 marks)
 - (ii) Explain the layout pane in JavaFX. (2 marks)
- (b) How do you handle user events button clicks in JavaFX, so that when the button is clicked, the message "Button clicked!" is printed to the console. Fill in the missing codes in **Figure 1** (5 marks)

```
public class ButtonClickExample extends Application
{
    @Override
    public void start(Stage primaryStage) {

        // Create a Button

        // Define an event handler for button click

        // Create a layout pane and add the button to it

        Scene scene = new Scene(root, 300, 250);
        primaryStage.setScene(scene);
        primaryStage.setTitle("Button Click Example");
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Figure 1 Codes that produce the output "Button clicked!" to the console.

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q1. (Continued)**

- (c) Create a simple JavaFX application that uses four buttons to move a rectangle as shown in the **Figure 2(a)**. Move the rectangle in the direction as indicated by the name of the button. Every click will move the rectangle by 2 pixels.

Complete the program in **Figure 2(b)** by filling out the missing codes for parts: (i), (ii), (iii), and (iv). You may omit the **import** statement in your answer.

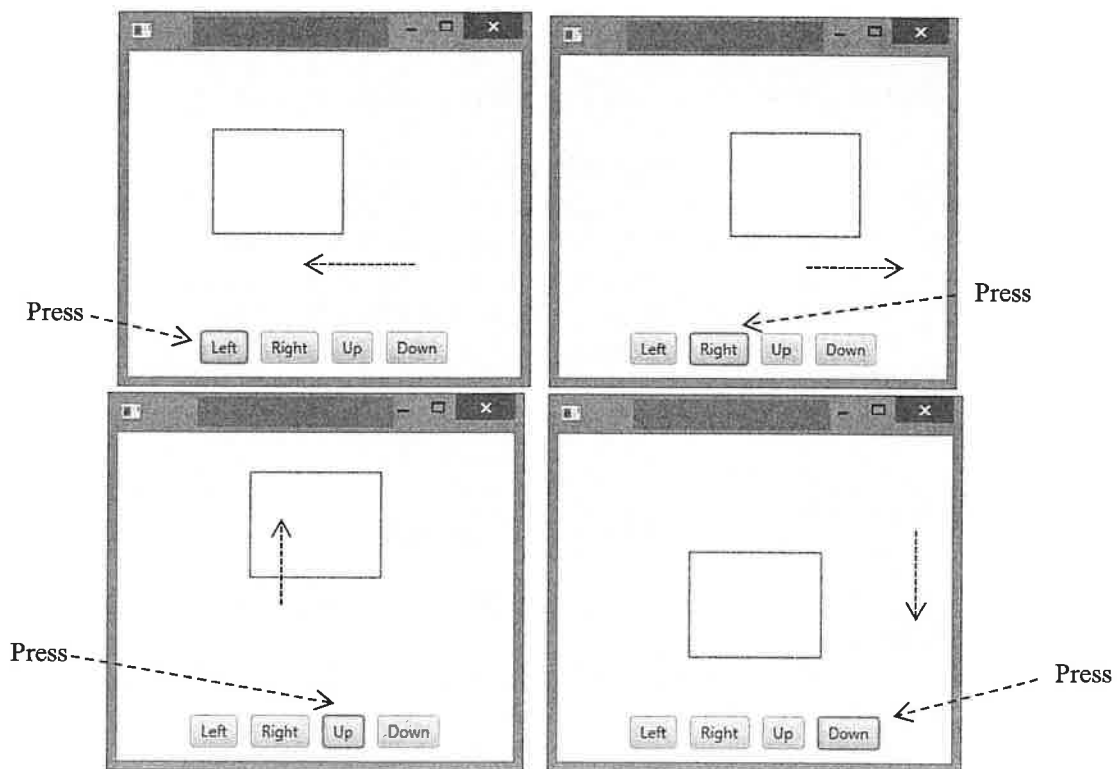


Figure 2(a) Example application with four buttons to change the direction of the rectangle

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q1. (c) (Continued)**

```
public class Question 1(c) extends Application {  
    public void start(Stage primaryStage) {  
        // Pane for placing the rectangle  
  
        // refer to Q1(c) (i)  
  
        // HBox pane for holding the four buttons  
        HBox hBox = new HBox();  
        hBox.setSpacing(10);  
        hBox.setAlignment(Pos.CENTER);  
        Button btLeft = new Button("Left");  
        Button btRight = new Button("Right");  
        Button btUp = new Button("Up");  
        Button btDown = new Button("Down");  
  
        hBox.getChildren().addAll(btLeft, btRight, btUp, btDown);  
  
        // refer to Q1(c) (ii)  
  
        // Create a border pane and place pane and hBox in the  
        border pane  
        BorderPane borderPane = new BorderPane();  
        borderPane.setCenter(pane);  
        borderPane.setPadding(new Insets(10));  
        borderPane.setBottom(hBox);  
        BorderPane.setAlignment(hBox, Pos.CENTER);  
  
        // refer to Q1(c) (iii)  
  
    }  
  
    // refer to Q1(c) (iv)  
}
```

Figure 2(b) Program to generate the application in **Figure 2(a)**

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q1. (c) (Continued)**

- (i) Add a Rectangle shape with size (100,80) at location (90,50) to pane. Set the stroke of the rectangle to red and fill the rectangle with white colour. (3 marks)
 - (ii) Add the event handling code for the 4 buttons using *Lambda* expression. Move the rectangle in the direction as indicated by the name of the button by 2 pixels when the button is pressed. Perform the respective actions as mentioned above (refer to **Figure 2(a)**) when these buttons are pressed. (6 marks)
 - (iii) Create a scene of size (300,250) to host the *pane*, place the scene into the stage and show the stage. (2 marks)
 - (iv) Add `main()` to launch the application. (1 mark)
- [Total : 25 marks]

Q2. (a) The following questions are based on File Input and Output,

- (i) Suppose you wanted to write code that reads from a file one word at a time. The code needs to peek ahead to find where the words are separated by whitespace. What input stream could you use to accomplish this and give a reason? (3 marks)
- (ii) What is the difference between the character streams and byte streams? (2 marks)

- (b) Write a simple Java program that writes the contents of a double array below to the binary file named "Number.dat". The binary file contains values in double format. You can omit the **import** statement in your coding.

(1.4, 2.6, 3.7, 4.25, 5.5, 6.8, 10.0) (7 marks)

(c) The following question is based on Java Database Programming.

- (i) How do you create a Statement and use it to update and query the database? (3 marks)
- (ii) How do you find the number of columns and columns names in the ResultSet? (3 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q2. (c) (Continued)**

- (iii) You are developing a Java application that needs to interact with a relational database. Explain how you would connect to a MySQL database, retrieve data from a table named "Employees", and display the employee id and name results in your application. (7 marks)

[Total : 25 marks]

- Q3. (a) Consider the following code snippet. The `checkCountry(String)` method throws an `InvalidCountryException` if the String given is not an actual country name.

```
Scanner input = new Scanner(System.in);
String defaultName = "Javaland";
String country;
try {
    System.out.println("Which country are you from?");
    country = input.next();
    checkCountry(country);
    System.out.print("Oh, you are from ");
}
catch (InvalidCountryException e) {
    System.out.println(country + " is not a country!");
    System.out.print("Then, you are from ");
    country = defaultName;
}
finally {
    System.out.println(country);
}
```

- (i) If the user enters "Kampar" (which is not a country), write all the text outputs in the order they are printed on the console screen. (3 marks)
- (ii) If the user enters "Malaysia" (which is a country), write all the text outputs in the order they are printed on the console screen. (2 marks)
- (iii) Write the class definition for `InvalidCountryException` exception, which should return "Invalid Country!" by its `getMessage()` method. (3 marks)
- (iv) What are the differences between checked and unchecked exceptions? Is `InvalidCountryException` a checked or unchecked exception? (3 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q3. (Continued)**

- (b) Consider the following code snippet.

```
public class ThreadTest {

    public static void main(String [] args) {
        PrintChar pa = new PrintChar('A');
        PrintChar pb = new PrintChar('B');

        // refer to (ii)

    }

    public class PrintChar // refer to (i)
    {
        private char c;
        public PrintChar(char c) {
            this.c = c;
        }
        public void run() {
            System.out.print(c);
        }
    }
}
```

- (i) Insert the code to make PrintChar a runnable task in a thread. (2 marks)
- (ii) Create one thread for pa and one thread for pb and start the two threads. (4 marks)
- (iii) Show all the possible outputs on the console screen when ThreadTest is executed. (2 marks)
- (c) What is a *race condition* in a multithreaded program? (2 marks)
- (d) If `xMethod()` causes race condition, provide and show two approaches in Java to avoid it. (4 marks)

[Total : 25 marks]

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**SECTION B**

This section consists of TWO (2) questions. Answer **ONLY ONE (1)** question.

Q4. A class Car is partially defined as follows:

```
abstract class Car
{
    private String model;
    private int year;

    public Car(String model, int year) {
        // refer to part(a)
    }

    public String getModel() {
        return model;
    }

    public int getYear() {
        return year;
    }

    public abstract boolean isElectric();
}
```

- (a) Write the statements for the constructor to initialize data member `model` and `year`. (2 marks)
- (b) This Car class is declared as abstract. What is an abstract class? (2 marks)
- (c) The `isElectric()` method is declared as abstract. What does this imply? (2 marks)
- (d) Honda is a concrete subclass of Car, which consists of an additional data member `engine` (String) representing the type of gasoline engine used. Write the class definition for Honda. The class should include a constructor for initializing all data members and the accessor for its data member. Override `isElectric()` to return *false* since it is not an electric car. (7 marks)
- (e) Tesla is another concrete subclass of Car. Write the class definition for Tesla. The class should include a constructor for initializing all data members. Override `isElectric()` to return *true* since it is an electric car. (4 marks)

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q4. (Continued)**

- (f) Suppose an application uses an array of a mixture of Honda and Tesla cars.
- (i) Write the statement to create an array of TEN (10) Car named *cars*. (2 marks)
 - (ii) Create a Honda car of model "Accord" and year 2024 with a "Turbo" engine. Assign the car to the first element of the *cars* array. (1 mark)
 - (iii) Create a Tesla car of model "Y" and year 2023. Assign the car to the second element of the *cars* array. (1 mark)
 - (iv) Assuming all the ten elements in the *cars* array have been initialized. Write a *for* loop to check and print "Electric" if the car is electric and print "Not Electric" if it is not. (4 marks)

[Total : 25 marks]

- Q5. (a) Given code snippet as below:

```
public class Q5 {
    public static void main(String[] args) {
        A a = new B();
        B b = new B();

        a.show(10, 10L);      // output 1
        a.show(10, 10.0);    // output 2
        b.show(10L, 10L);    // output 3
        b.show(10, 10.0);    // output 4
    }
}

class A {
    void show(int i, double j) {
        System.out.println("A : int, double");
    }

    void show(int i, long j) {
        System.out.println("A : int, long");
    }
}

class B extends A {
    void show(int i, long j) {
        System.out.println("B : int, long");
    }

    void show(long i, long j) {
        System.out.println("B : long, long");
    }
}
```

UCCD2044 OBJECT-ORIENTED PROGRAMMING PRACTICES**Q5. (a) (Continued)**

- (i) Show the outputs when the `main()` method is executed. (4 marks)
 - (ii) What is the difference between overloading and overriding? (4 marks)
 - (iii) Of the FIRST TWO (2) outputs in (i), indicate whether they are the results of overloading or overriding. (2 marks)
- (b) Describe the three principles of object-orientated programming? (6 marks)
- (c) Are the following two classes immutable? Justify your answers. (4 marks)

<pre>public class A { private int[] value; public int[] getValue(){ return value; } }</pre>	<pre>public class B { private String value; public String getValue(){ return value; } }</pre>
--	--

- (d) Write a method that finds the number of occurrences of a specified character (char a) regardless of case in the string (String str) using the following header.

```
public static int count(String str, char a)
```

For example, `count("Evening", 'e')` returns 2.

(5 marks)

[Total : 25 marks]